BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No. : **BML/QA/APQR/QA/002**

Protocol Date : **23/04/2024**

Total No. of Pages :

PRODUCT NAME : Bioflex Dextrose 50%w/v Intravenous Injection B.P 100ml

GENERIC NAME : Dextrose 50% w/v Intravenous Injection B.P 100ml

COMPOSITION : Each 100ml contains
Dextrose 50g
Water for injection B.P q.s

PACK SIZE : 100ml in a printed PVC pouch

PACKAGING TYPES AVAILABLE : 100ml Polyvinylchloride PVC pouch
Corrugated cardboard boxes carton

MASTER FORMULAR NUMBER : BML/QA/MFR/016

SHELF LIFE: 3 Years

TIME PERIOD COVERED : JANUARY – DECEMBER, 2025

PRODUCT CODE : 50G

BIOMEDICAL LIMITED

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Sr. No	Subject	Name	Designation	Sign/Date
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4	Authorized by	Agboola L. A.	Factory Manager	

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1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay) and microbiological analysis (Sterility and Pyrogen), post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Bioflex Dextrose 50%w/v Intravenous Injection is a clear colorless sterile solution in a PVC pouch with double port for easy administration. Formulated for the treatment of insulin hypoglycemia to restore glucose levels, the hypertonic solution is carefully controlled to ensure the glucose level of 47.5-52.5g/100ml. Careful control of all other parameters as mentioned in the scope and under the section for the review of the quality control of the product in this piece are always assured.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Bioflex Dextrose 50%w/v Intravenous Injection comprises of about 11,850 bags. Total No. of Batches produced within the period January 2025 to December 2025 is 15 batches i.e. from Batch no.:50G25001 to 50G25015; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Bioflex Dextrose 50%w/v Intravenous Injection are Dextrose monohydrate (active ingredient) and Water for injection (excipient). Dextrose monohydrate is sourced from approved vendors while water for injection is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table below shows the different consignments of dextrose monohydrate received and used in the manufacture of Bioflex Dextrose 50%w/v Intravenous Injection within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

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3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only (see BML/QA/AVL/001).

3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Dextrose Monohydrate	Weifang Shengtai Medicine Co. Ltd. China	Weifang Shengtai Medicine Co. Ltd. China	**7**	**7**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**1**	**1**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Kuber, Fzco Dubai Silicon, Oasis.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Farmasino Co.Ltd China.	**2**	**2**	**Nil**	**Nil**
	Weifang Shengtai Medicine Co. Ltd. China	Bdv Behrens GmbH Hamburg, Germany.	**2**	**2**	**Nil**	**Nil**

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3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in poly vinyl chloride (PVC) self-collapsible soft bag (pouch) bought as a finished product from the manufacturer of the PVC pouch. Manufacturer COA of the PVC is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers as conform to in-house specifications set for the primary packaging material.

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
PVC pouches	Jiangsu Medwell Medical Equipment Co. Ltd. China	Jiangsu Medwell Medical Equipment Co. Ltd. China	16	16	Nil	Nil

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Clear, colorless and particles free solution
Dextrose B.P.: 95 – 105%
pH: 3.20 – 6.50
Hydroxymethylfurfural (HMF): NMT 0.25

3.3.2 IN-PROCESS RESULTS:

The summary of in-process results of all the batches manufactured was prepared and the table showing the results presented. The summary of in-process results for the chemical analysis and microbiological analysis of the batches manufactured is shown

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Chemical Analysis of IN-PROCESS Result

S/N	**BATCH NO**	**Dextrose** Assay (95 – 105%)	**pH** (sample) (3.2 – 6.5)	**pH** (WFI) (5.0 – 7.0)
1	50G25001	97.68	4.88	6.14
2	50G25002	96.9	5.21	6.18
3	50G25003	97.64	5.3	6.1
4	50G25004	96.03	5.72	6.48
5	50G25005	98.58	5.12	6.1
6	50G25006	100.3	5.1	6.16
7	50G25007	97.49	5.3	6.51
8	50G25008	98.91	5.32	5.93
9	50G25009	96.06	5.17	6.03
10	50G25010	98.95	5.54	6.12
11	50G25011	98.9	5.69	6.32
12	50G25012	96.53	5.1	6.18
13	50G25013	98.57	5.17	6.35
14	50G25014	97.71	5.6	6.42
15	50G25015	98.75	4.94	6.05

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Chemical/Microbial Analysis for FINISHED PRODUCT Result

S/N	**BATCH NO**	**Dextrose** Assay (95 – 105%)	**pH** (3.2 – 6.5)	**HMF** (NMT0.25)	**Sterility** (MUST BE STERILE)	**BACTERIAL ENDOTOXINS**
1	50G25001	98.31	3.36	0.081	STERILE	<0.125
2	50G25002	98.62	3.53	0.077	STERILE	0.5
3	50G25003	98.47	3.72	0.085	STERILE	<0.125
4	50G25004	98.62	3.55	0.083	STERILE	<0.125
5	50G25005	97.89	3.92	0.08	STERILE	<0.125
6	50G25006	96.55	3.55	0.075	STERILE	0.15
7	50G25007	96.29	3.67	0.082	STERILE	0.4
8	50G25008	98.5	3.49	0.084	STERILE	<0.125
9	50G25009	96.18	3.62	0.087	STERILE	0.2
10	50G25010	97.03	3.73	0.082	STERILE	0.2
11	50G25011	97.54	3.48	0.075	STERILE	<0.125
12	50G25012	98.6	3.54	0.082	STERILE	<0.125
13	50G25013	97.18	3.71	0.081	STERILE	<0.125
14	50G25014	98	3.3	0.081	STERILE	0.05
15	50G25015	98.04	3.44	0.076	STERILE	<0.125

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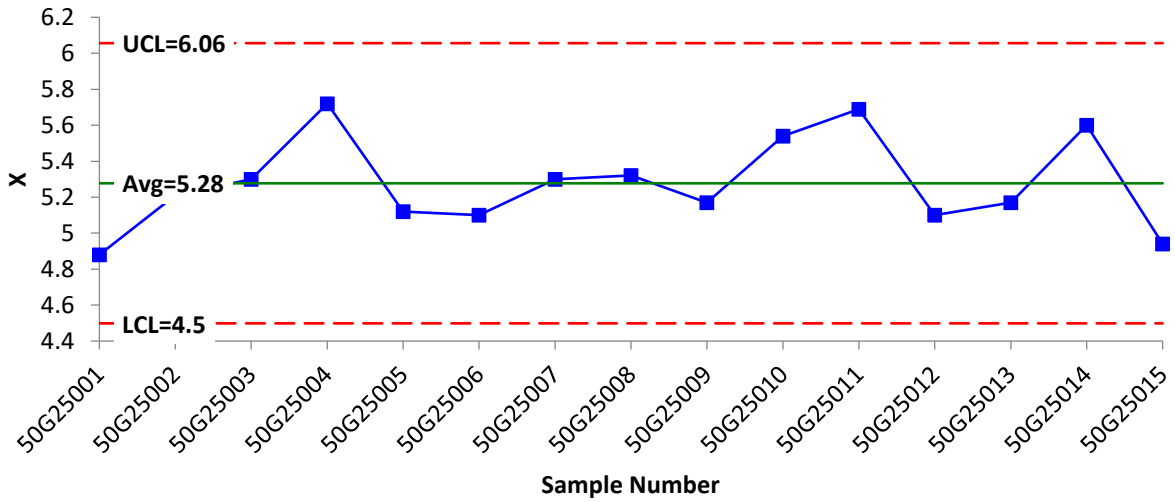
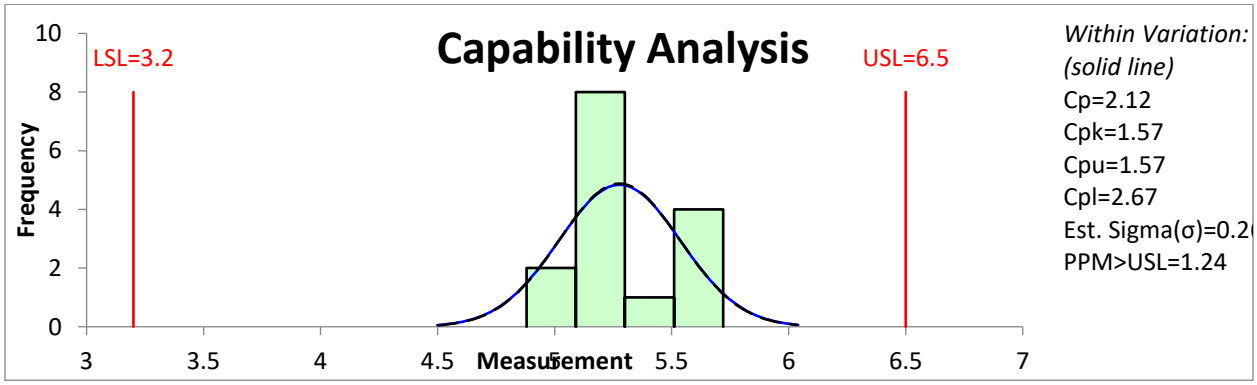
3.3.3 TREND ANALYSIS

Trend analysis of Dextrose content, pH, HMF, Pyrogen and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for inprocess analysis of the intermediate

Specification (3.2 -6.5**)**





Mean	5.28
Standard deviation	0.257611742
Maximum	5.72
Minimum	4.88
Control limit: 4.50– 6.06	
Alert limit: 4.76 – 5.80	

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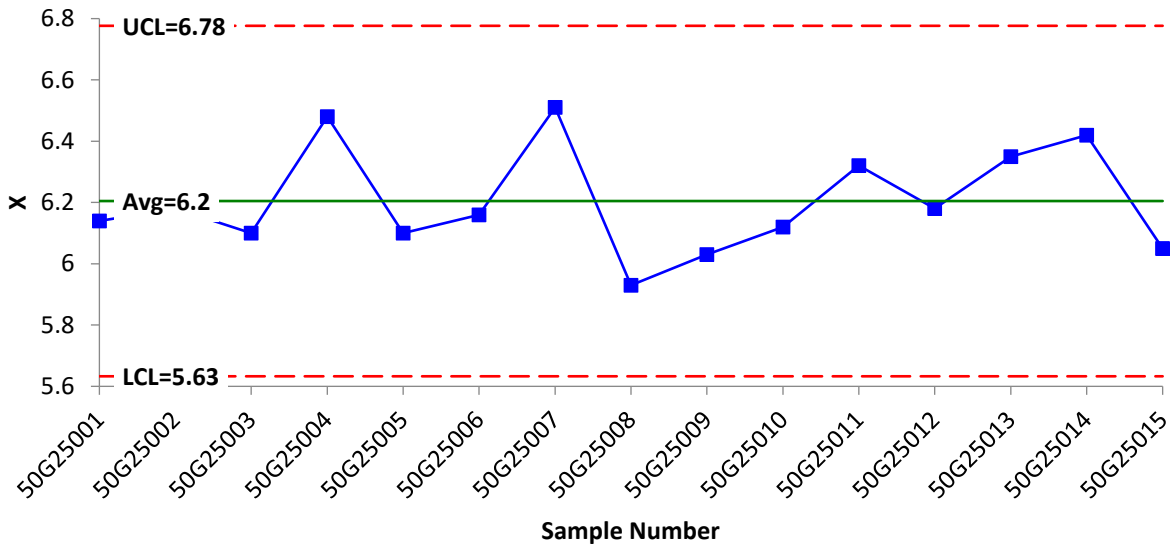
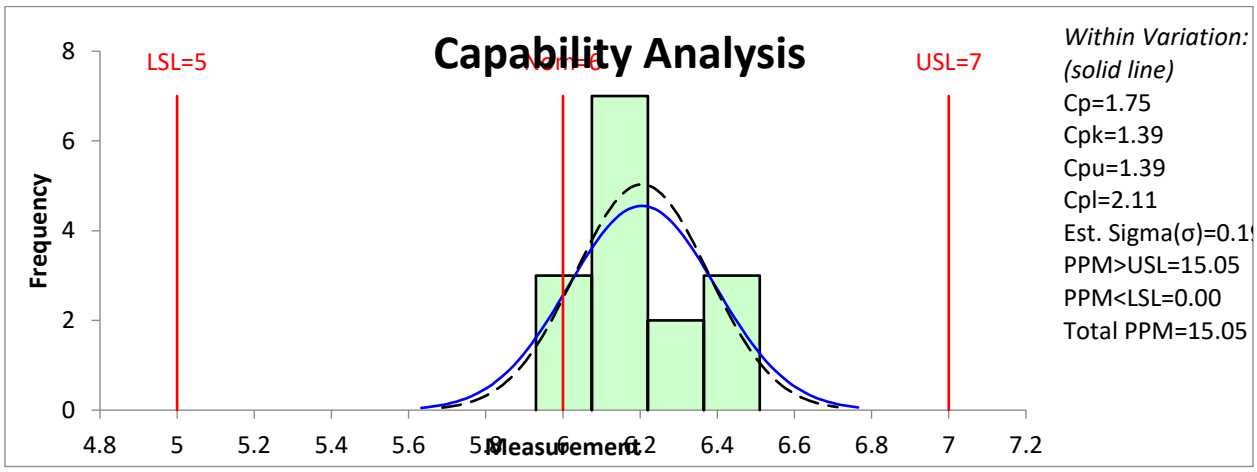
Minimum-Maximum: 4.88–5.72
Alert limit minimum well within the specification limit, control limit existing within the specification limit.

Annexure II: pH trend for Water for Injection (WFI)

Specification (5.0 -7.0**)**





Mean	6.2
Standard deviation	0.17241423
Maximum	6.51
Minimum	5.93
Control limit: 5.63– 6.78	
Alert limit: 5.86 –6.54	

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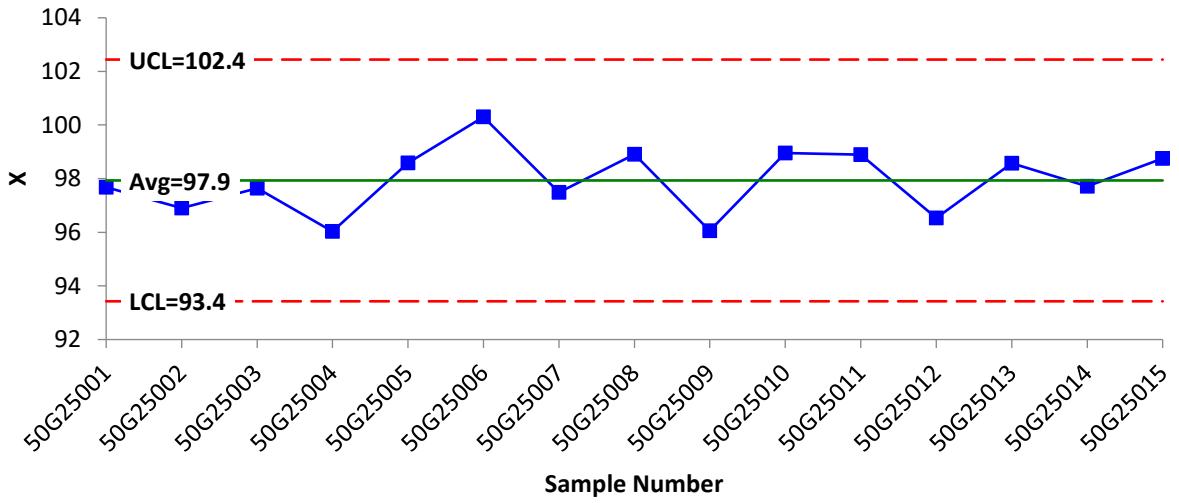
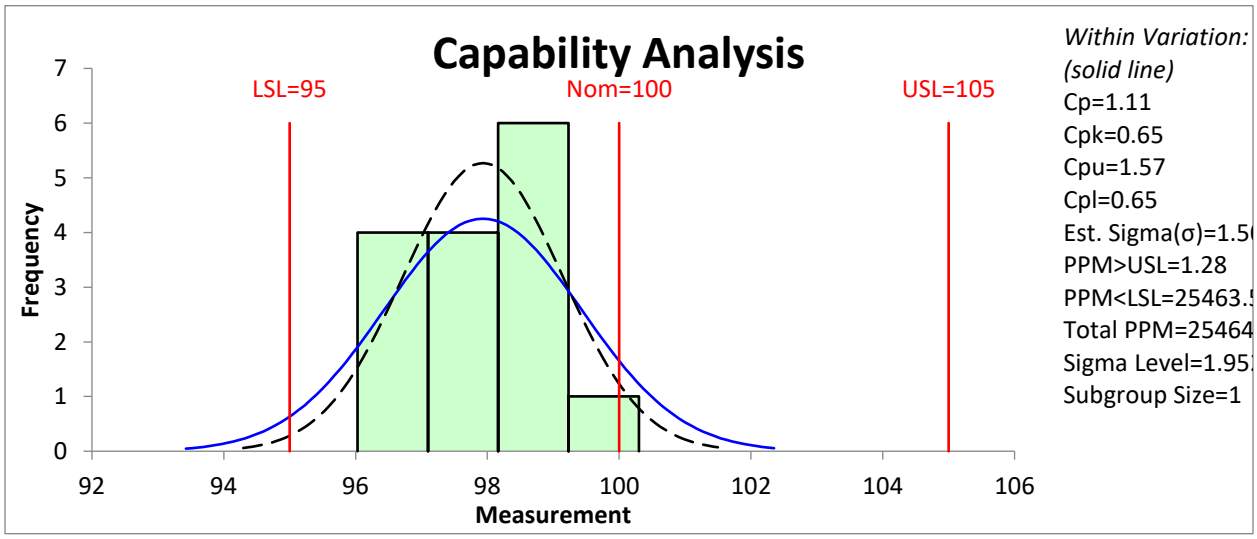
Minimum-Maximum: 5.93–6.51
Alert limit minimum well within the specification limit, control limit existing within the specification limit.

Annexure III: Assay trend of in-process analysis of Dextrose in Dextrose 50% Injection

Specification of Dextrose (95 – 105%**)**





Mean	97.9
Standard deviation	1.212415927
Maximum	100.3
Minimum	96.03
Control limit: 93.4–102.4	

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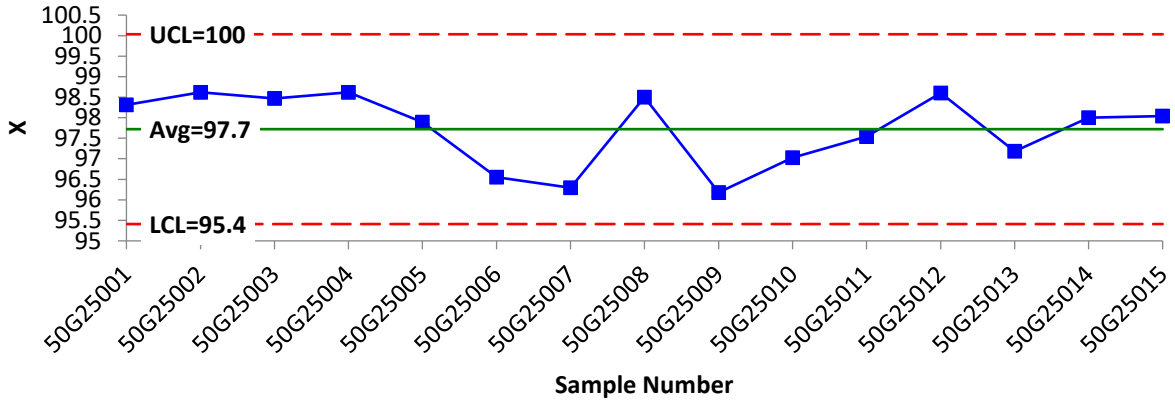
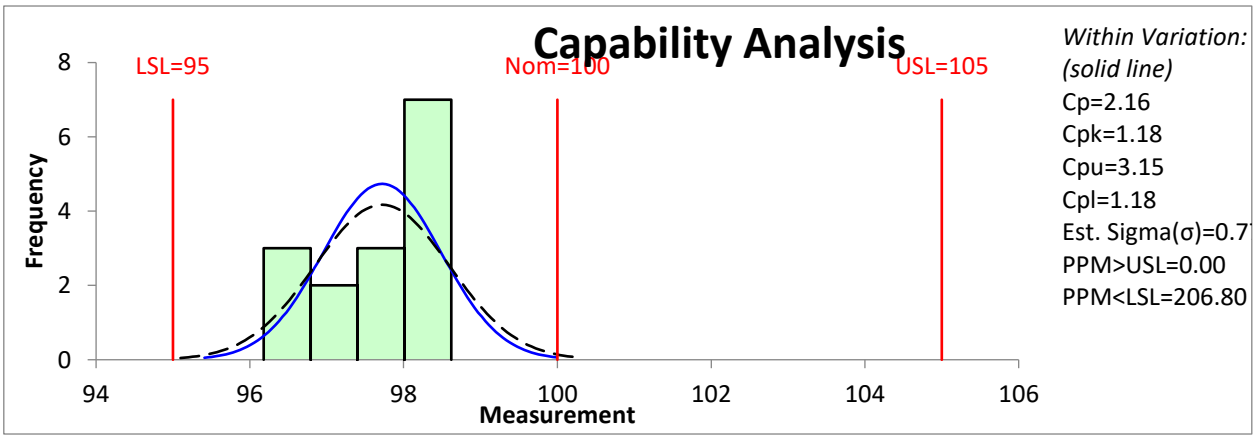
Alert limit: 95.48 –100.32
'Minimum-Maximum: 96.03 –100.3
Alert limit minimum within the specification limit.

Annexure IV: Assay trend of finished product analysis of Dextrose in Dextrose 50% Injection

Specification of Dextrose (95 – 105%**)**





Mean	97.7
Standard deviation	0.874584527
Maximum	98.62
Minimum	96.18
Control limit: 95.4 –100	
Alert limit: 95.95 –99.45	
Minimum-Maximum: 96.18–98.62	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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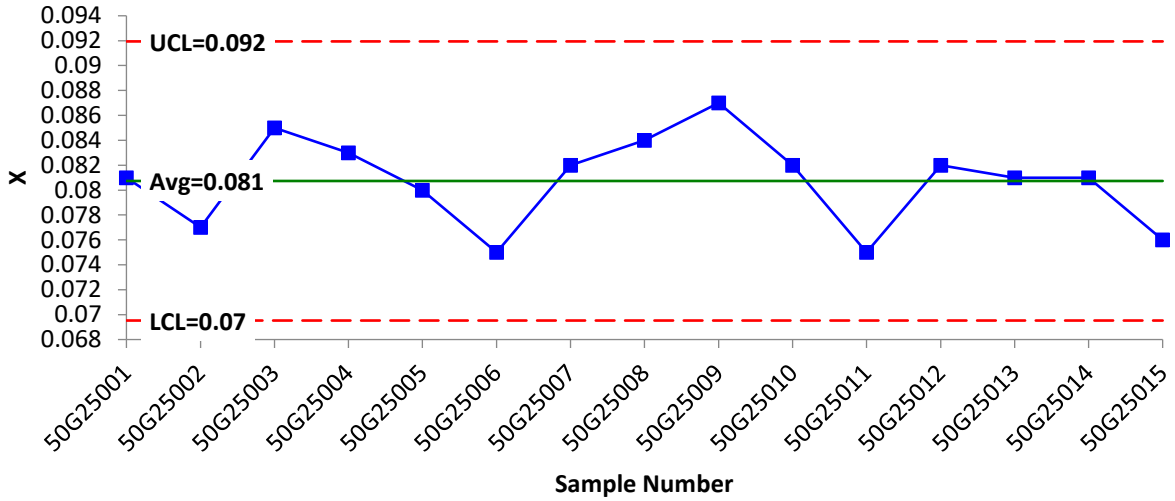
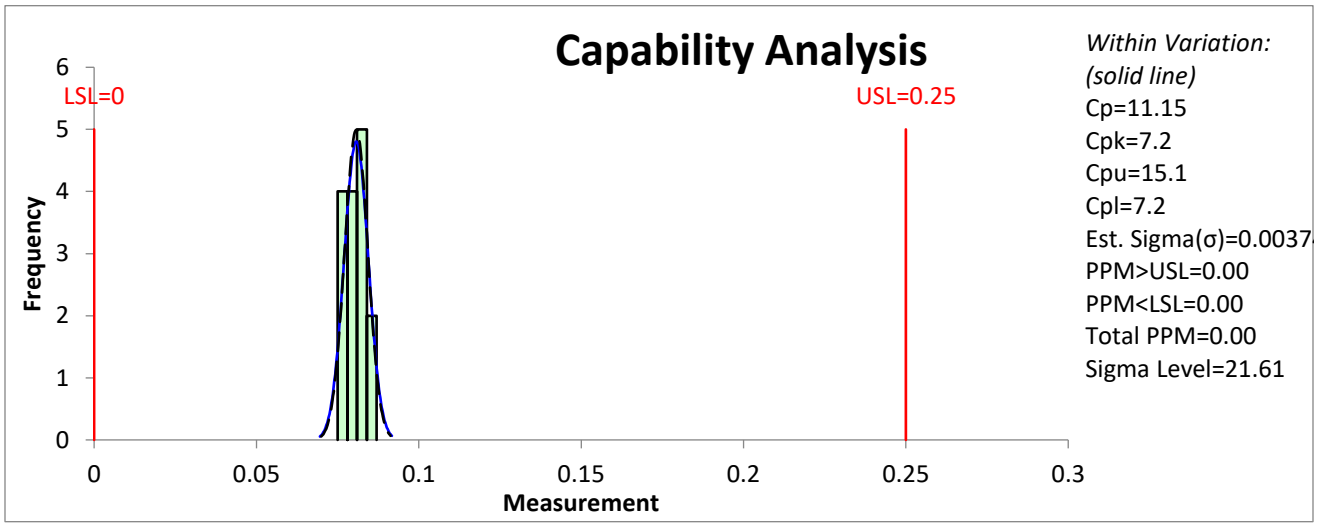
Annexure V: Assay trend of finished product analysis of HMF in Dextrose 50% Injection

Specification of Dextrose (95 – 105%)





Mean	0.081
Standard deviation	0.003595
Maximum	0.087
Minimum	0.075
Control limit: 0.07–0.092	

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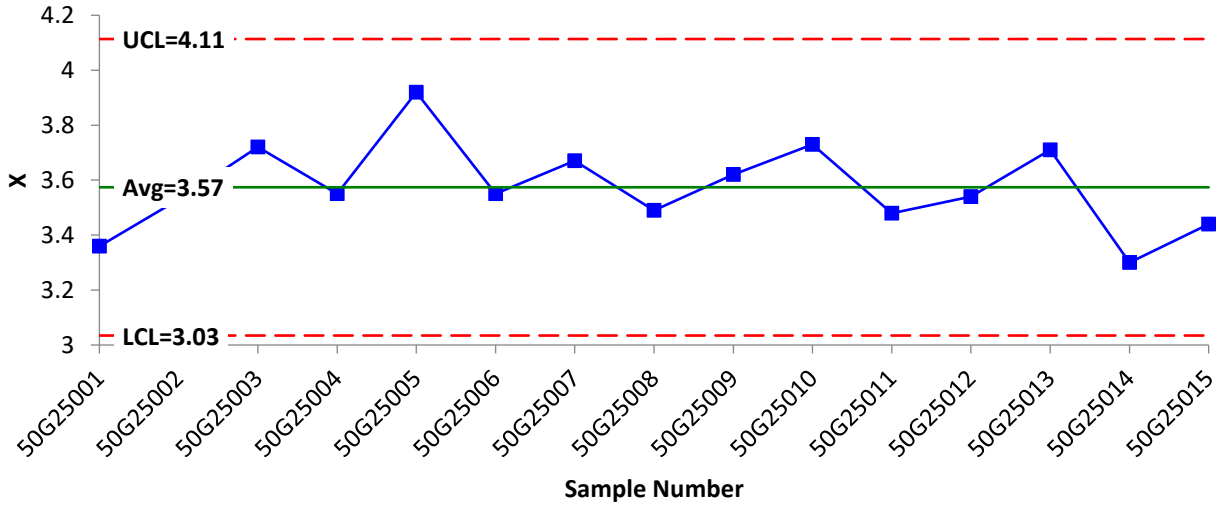
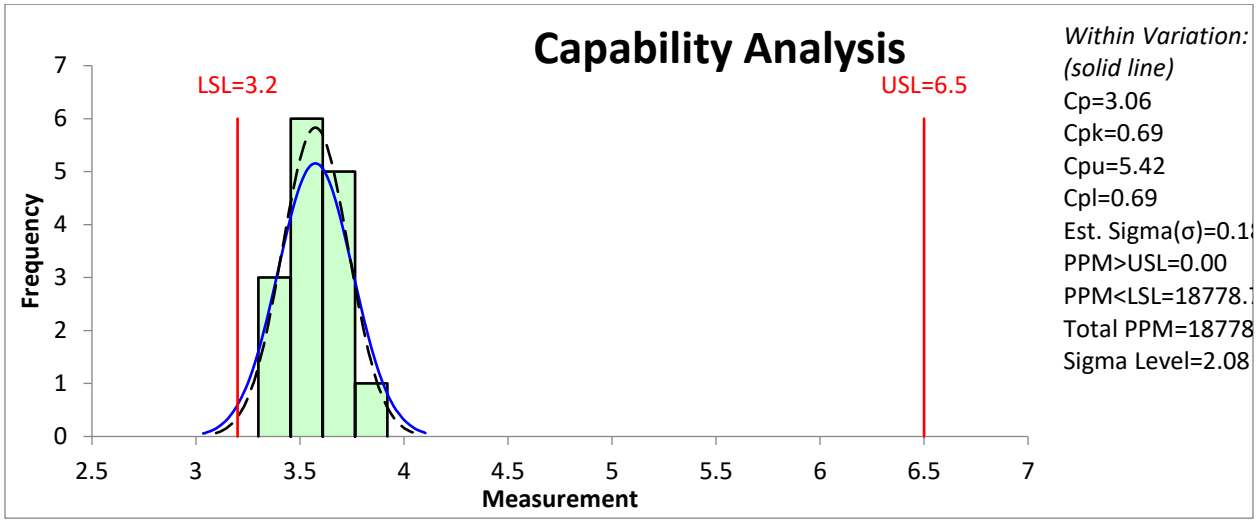
Alert limit: 0.07381 – 0.08819
Minimum-Maximum: 0.075–0.087
Alert limit within the specification limit, control limit existing within the specification limit.

Annexure VI: pH trend of finished product analysis of Dextrose 50% Injection

Specification of pH (3.20-6.50**)**





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Mean	3.57
Standard deviation	0.159050756
Maximum	3.92
Minimum	3.3
Control limit: 3.03 – 4.11	
Alert limit: 3.25–3.89	
Minimum-Maximum: 3.3–3.92	
Alert limit within the specification limit, control limit existing within the specification limit.	

SUMMARY OF PROCESS CAPABILITY

PROCESS CAPABILITY FOR INTERMEDIATE		
1	pH	1.57
2	Rinse Water	1.39
3	Dextrose	0.65
PROCESS CAPABILITY FOR PRODUCT		
1	pH	0.69
2	Dextrose	1.18
3	HMF	7.2

CQAs	Intermediate		Finished product	
	Control limit	Process capability	Process capability	Control limit
pH	pass	1.57	0.69	fail
HMF	-	-	7.2	pass
Dextrose content	fail	0.65	1.18	pass

Two parameters — finished product pH (Cpk 0.69) and in-process Dextrose assay (Cpk 0.65) returned capability indices below 1.00 and will be subject to formal root cause investigation and enhanced monitoring in 2026. The inverse capability trend observed between stages for both parameters (pH improving at intermediate, declining at finished product; Dextrose declining at intermediate, recovering at finished product) provides important directional information for focused investigation. No out-of-specification results were recorded across all 15 batches during the review period, confirming that the

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process remains in a state of control. All corrective and preventive actions arising from capability investigations will be tracked under the CAPA system and reviewed at the next APQR cycle.

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

No observed deviation from the predetermined tolerance was observed.

3.5.3 NON CONFORMANCE

No observed non-conformance from the predetermined tolerance was observed.

3.5.4 BATCH FAILURES

There was no batch failure observed during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Dextrose 50% Intravenous Injection during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form.

3.7 STABILITY MONITORING PROGRAMME

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The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Appearance, Pyrogenicity and Sterility are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2025 to Dec. 2025.

3.8.2 RETURNED GOODS FROM MARKET

No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

3.8.4 SALVAGED GOODS

There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

3.10.1.1 MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Solution preparation
Batch coding of PVC bag (packaging material)
Solution filling
Sterilization of the filled solution in qualified autoclave
Final packaging of the sterilized solutions

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ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay
Chemical testing of the finished products
Microbiological tests on the finished product-
Sterility tests
Bacterial endotoxin tests (Pyrogen and Limulus Amoebocyte Lysate)

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 8 batches of Dextrose 50% Intravenous Injection 100ml with same manufacturing formula since Jan., 2025 till Dec., 2025 and yearly APQR data shows satisfactory trend throughout the period. Also the retained samples prove that the product manufactured with the existing formulation is stable.

3.10.1.2 ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit and batch coding unit

Environmental Conditions
Std Temp. : NMT 30°C
Relative Humidity: NMT 65%

3.10.1.3 CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.12 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no

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report of product contamination nor pyrogenicity has been reported with the product within the year under review.

3.14 COMPARATIVE PRODUCT QUALITY REVIEW

CONCLUSION

The process capability indices (Cpk) were calculated for all critical quality attributes monitored during in-process and finished product testing across the 15 batches manufactured in 2025. The results are interpreted as follows:

In-Process Parameters

The pH of the intermediate product returned a Cpk of **1.57**, indicating an excellent, well-centred process with substantial margin from specification limits. This reflects consistent WFI quality and reliable pH adjustment during compounding. The WFI rinse water pH similarly returned a Cpk of **1.39**, confirming stable and controlled water system performance throughout the year.

The in-process Dextrose assay, however, returned a Cpk of **0.65**, which falls below the universally accepted minimum capability threshold of 1.00. While all 15 batches remained within specification (95–105%), this low index indicates that the process is operating closer to one of its specification boundaries than is desirable, and the current process spread relative to the specification window leaves limited tolerance for natural variation. This parameter will be placed under enhanced monitoring in 2026, and a formal investigation will be initiated to identify and address root causes — including potential contributions from dispensing accuracy, balance calibration, analytical method variability, and sampling technique.

Finished Product Parameters

The finished product Dextrose assay returned a Cpk of **1.18**, which is acceptable and reflects improved process control at the final product stage relative to the in-process result. This is consistent with the narrower result range observed (96.18–98.62%) and confirms that the sterilisation and filling steps do not materially degrade assay performance.

The finished product pH returned a Cpk of **0.69**, again below the 1.00 threshold. All 15 batches passed the specification limit of 3.20–6.50; however, the statistical spread of results (3.30–3.92), combined with the relatively tight lower end of the distribution, warrants attention. The low Cpk is partly attributable to the very wide specification range producing a mean (3.57) that sits closer to the lower boundary than the centre. A review of pH adjustment steps and the impact of sterilisation on final product pH will be conducted in Q1 2026.

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The HMF content returned an outstanding Cpk of **7.20**, confirming that sterilisation cycle parameters (time, temperature, and F_o delivery) are well controlled and that thermal degradation of dextrose is minimal and consistently far from the NMT 0.25 limit. This is a strong indicator of autoclave cycle reproducibility.

Overall Capability Assessment

With the exception of in-process Dextrose assay and finished product pH — both of which will be subject to enhanced monitoring and root cause investigation — all other parameters demonstrate satisfactory to excellent process capability. No out-of-specification results were recorded for any batch during the review period, confirming that the process, while capable of improvement in two areas, remains in a state of control. Corrective actions arising from capability investigations will be tracked under the CAPA system and reviewed at the next APQR cycle.

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